

Bellman Equation Using the MDP

- System of linear equations; one for each state s

$$\begin{aligned} v_{\pi}(s) &= E[R_{t+1} + \gamma v_{\pi}(S_{t+1}) | S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s', r} (r + \gamma v_{\pi}(s')) p(s', r | s, a) \end{aligned}$$

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$$\begin{aligned} v_{\pi}(s) &= E[R_{t+1} + \gamma v_{\pi}(S_{t+1}) | S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s', r} (r + \gamma v_{\pi}(s')) p(s', r | s, a) \end{aligned}$$

- System of linear equations; one for each state (s, a)

$$\begin{aligned} q_{\pi}(s, a) &= E[R_{t+1} + \gamma v_{\pi}(S_{t+1}) | S_t = s, A_t = a] \\ &= \sum_{s', r} (r + \gamma v_{\pi}(s')) p(s', r | s, a) \end{aligned}$$

Optimal State and Action Value Functions

$$v_*(s) = \max_{\pi} v_{\pi}(s), s \in S$$

$$q_*(s, a) = \max_{\pi} q_{\pi}(s, a), \forall s, a$$

Optimal State and Action Value Functions

$V_*(s) \geq V_\pi(s)$ for all policies π .

$$v_*(s) = \max_{\pi} v_{\pi}(s), s \in S$$

$$q_*(s, a) = \max_{\pi} q_{\pi}(s, a), \forall s, a$$

Over the set of all policies (functions) that map states to actions, look for policies that for every state s in the set of states S , maximize the state's value.

Bellman Optimality Equations

We can write:

$$v_*(s) = E_{\pi_*} [R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s]$$

Optimality principle:

- It is sufficient to pick the optimal action at time t (given $v_*(S_{t+1})$)

$$\rightarrow v_*(s) = \max_a E [R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s, A_t = a]$$

$$\max_a q_*(s, a)$$

$$S = \{s_1, s_2\}$$

$$\pi_*^{(1)} \quad \pi_*^{(2)}$$

$$V_{\pi_*^{(1)}}(s_1) > V_{\pi_*^{(2)}}(s_1)$$

$$V_{\pi_*^{(1)}}(s_2) < V_{\pi_*^{(2)}}(s_2)$$

Neither policy is optimal. You could improve $\pi_*^{(1)}$ by setting $\pi_*^{(1)}(s_2) = \pi_*^{(2)}(s_2)$. $\pi_*^{(1)}$ now results in state values at least as large as $\pi_*^{(2)}$.

Bellman Optimality Equations

We can write:

$$v_*(s) = E_{\pi^*} [R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s]$$

Optimality principle:

- It is sufficient to pick the optimal action at time t

$$v_*(s) = \max_{a \in A(s)} E[R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s, A_t = a], \forall s \in S$$

Bellman Optimality Equations

$$v_*(s) = \max_{a \in A(s)} E[R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s, A_t = a], \forall s \in S$$

What are we averaging over?

- We have one such equation for every state s
- How many equations and unknowns?

Bellman Optimality Equations: Expand the E[]

- A nonlinear system of equations
- For any state s

$$\begin{aligned}
 v_*(s) &= \max_{a \in A(s)} \overset{\downarrow}{E}[R_{t+1} + \gamma v_*(S_{t+1}) | \underbrace{S_t = s, A_t = a}_\phi], \\
 &= \max_{a \in A(s)} \sum_{r} \sum_{s'} (\underbrace{r + \gamma v_*(s')}_{\text{handwritten}}) \underbrace{P[R_t = r, S_{t+1} = s' | S_t = s, A_t = a]}_{\text{handwritten}}
 \end{aligned}$$

Bellman Optimality Equations

- A nonlinear system of equations
- For any state s

$$\begin{aligned}
 v_*(s) &= \max_{a \in A(s)} E[R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s, A_t = a], \\
 &= \max_{a \in A(s)} \sum_{s', r} (r + \gamma v_*(s')) p(s', r | s, a)
 \end{aligned}$$

Bellman Optimality Equations

- A nonlinear system of equations
- For any (s, a)

$$q_*(s, a) = E[R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s, A_t = a],$$

$$= \boxed{\max_{a' \in A(s_t)} q_*(S_{t+1}, a')}$$

Bellman Optimality Equations

- A nonlinear system of equations
- For any (s, a)

$$\begin{aligned} q_*(s, a) &= E[R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s, A_t = a], \\ &= \max_{a \in A(s)} \sum_{s', r} (r + \gamma \max_{a' \in A(s')} q_*(s', a')) p(s', r | s, a) \end{aligned}$$

Bellman Optimality Equations

- You are given the optimal **action**-value function. How will you calculate the optimal policy?

For every state s , you are given

$$q_*(s, a) \text{ for every } a \in A(s)$$

$$\pi_*(s) = \underset{a \in A(s)}{\operatorname{argmax}} q_*(s, a), \text{ for every } s.$$

Bellman Optimality Equations

- You are given the optimal **state**-value function. How will you calculate the optimal policy?

$$\pi_*(s) = \underset{a \in A(s)}{\operatorname{argmax}} \underbrace{E[R_{t+1} + \gamma V_*(S_{t+1}) | S_t = s, A_t = a]}_{q_*(s, a)}, \text{ for every } s.$$

The Dynamic Programming Algorithm

Bellman's Optimality Principle

Operations T_π and T

$$T_\pi V(s) = E_\pi [R_{t+1} + \gamma V(S_{t+1}) | S_t = s], \text{ for every state } s.$$

$$TV(s) = \max_{a \in A(s)} E [R_{t+1} + \gamma \underbrace{V(S_{t+1})}_{\text{any reward-to-go function } v(\cdot)} | S_t = s, A_t = a]$$

Bellman Equations Using Operators

Bellman's Equations Using T_π

$$V_\pi(s) = \underbrace{E_\pi \left[R_{t+1} + \gamma V_\pi(S_{t+1}) \mid S_t = s \right]}$$

$$= T_\pi V_\pi(s)$$

$$V_\pi(s) = T_\pi V(s),$$

for every state s .

Optimality Equations Using Operators

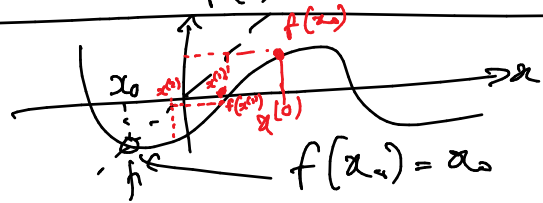
Bellman's Equations Using T

$$V_*(s) = \max_{a \in A(s)} E[R_{t+1} + \gamma V_*(S_{t+1}) \mid S_t = s, A_t = a]$$

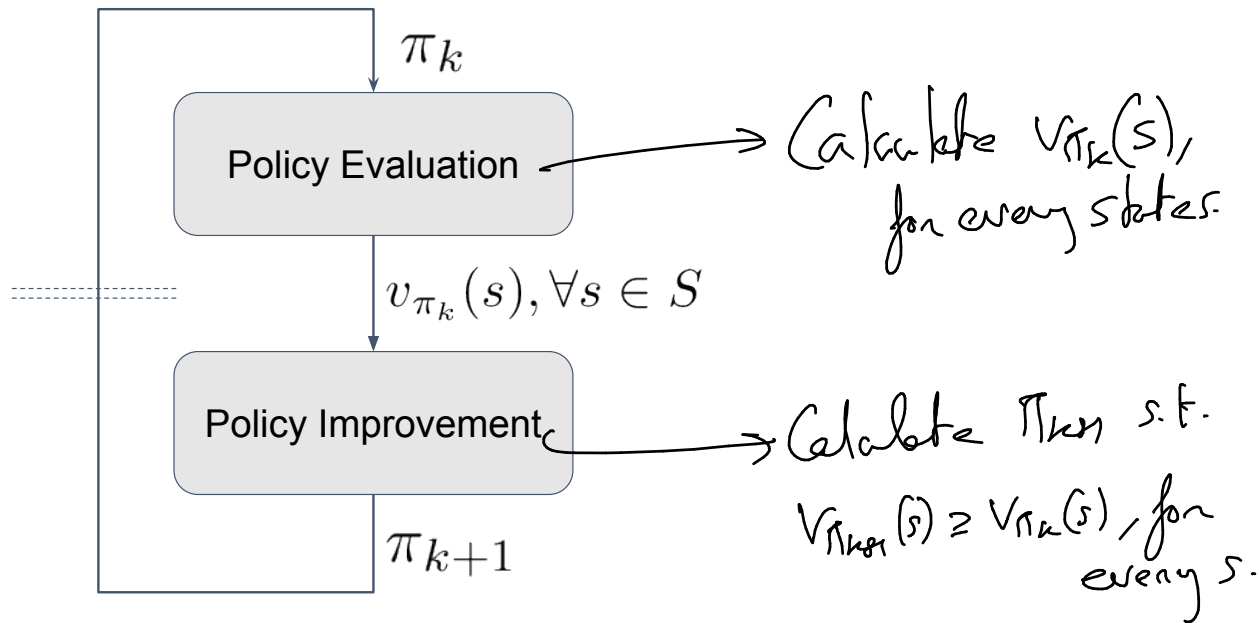
$$V_*(s) = T V_*(s) \quad \text{for every } s \in \mathcal{S}$$

$$V_*(s) = T V_*(s)$$

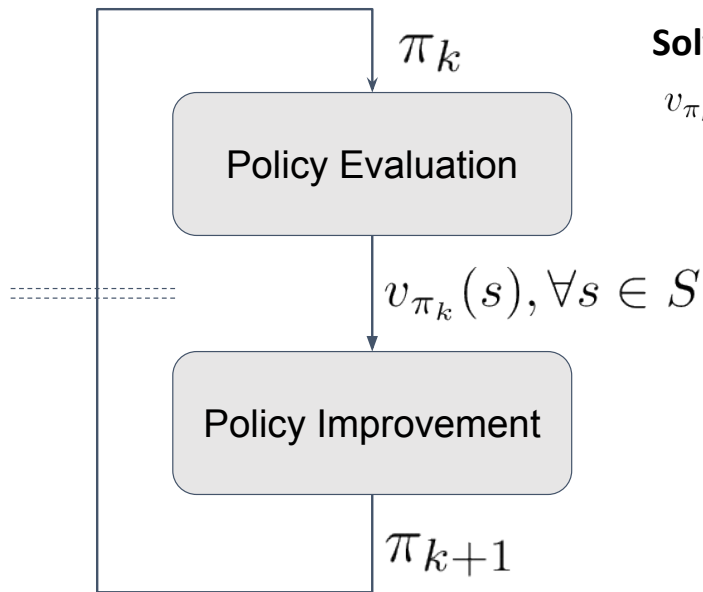
$$V_\pi(s) = T_\pi V_\pi(s)$$



Policy Iteration



Policy Iteration

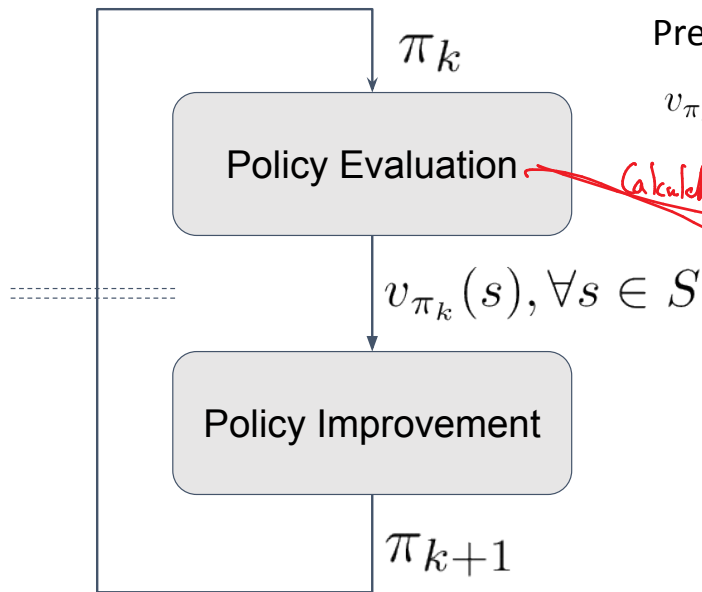


Solve:

$$\begin{aligned}
 v_{\pi_k}(s) &= E[R_{t+1} + \gamma v_{\pi_k}(S_{t+1}) | S_t = s] \\
 &= \sum_a \pi(a|s) \sum_{s', r} (r + \gamma v_{\pi_k}(s')) p(s', r | s, a)
 \end{aligned}$$

Policy Iteration

Exit condition:
 exit in case
 $\max_s |V_{\pi_k}^{(j)}(s) - V_{\pi_k}^{(j-1)}(s)| < \epsilon$
 $\epsilon > 0$



Prefer to Solve Iteratively:

$$v_{\pi_k}(s) = E[R_{t+1} + \gamma v_{\pi_k}(S_{t+1}) | S_t = s]$$

$$= \sum_a \pi_k(a|s) \sum_{s', r} (r + \gamma v_{\pi_k}(s')) p(s', r | s, a)$$

Calculate v_{π_k}
 Start with guess $V_{\pi_k}^{(0)}(s)$, for every s .
 while () {
 for s in S

$$\{ V_{\pi_k}^{(j)} = \sum_a \pi_k(a|s) \sum_{s', r} (r + \gamma V_{\pi_k}^{(j-1)}(s')) p(s', r | s, a) \}$$

}
 $j = j + 1$ } $\rightarrow$ [Exit cond]

Policy Evaluation Iterative Solution

- Exploit properties of the operator

